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From Atherosclerosis to Acute Coronary Syndromes: The Role of Soluble CD40 Ligand☆

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The binding of CD40 ligand (CD40L) to CD40 stimulates inflammatory processes including the release of proinflammatory cytokines and the expression of adhesion molecules implying a role in atherosclerosis. Patients exhibiting hypercholesterolemia, unstable angina, or acute myocardial infarction present with increased CD40L levels. Novel data suggest that elevated soluble CD40L levels not only represent a risk factor for cardiovascular disease but also predict future adverse events, especially in patients with acute coronary syndromes (ACS). Examination of the potential role of the genetic variability on CD40/CD40L genes in ACS, as regards the regulation of CD40L, appears to be of great interest. Moreover, several therapeutic approaches such as statins, antihypertensive agents, and antiplatelet agents have been suggested as potential modulators of CD40L levels anticipating a positive impact on the outcomes of patients with ACS. Whether specific agents target the CD40/CD40L system as well as its pathogenic role in ACS remains to be elucidated by large-scale studies in the future. (Trends Cardiovasc Med 2010;20:153–164) *© 2010, Elsevier Inc. All rights reserved.*

• Introduction

Disruption of vulnerable atheromatous plaque is the most common pathogenic mechanism in acute coronary syndromes (ACS). There is increasing evidence that inflammation plays a key role in the pathogenesis of atherosclerosis

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